

Checklist

What you should know

After studying this subtopic you should be able to:

- Deduce oxidation states of an atom in a compound or an ion.
- Identify the oxidised and reduced species and the oxidising and reducing agents in a chemical reaction.
- Deduce redox half-equations and equations in acidic or neutral solutions.
- Determine the relative ease of oxidation of different metals by analysing redox reactions between metals and aqueous metal ions.
- Predict the relative ease of oxidation of metals.
- Predict the relative ease of reduction of halogens.
- Interpret data regarding metal and metal ion reactions.
- Deduce equations for reactions of reactive metals with dilute HCl and H_2SO_4 .
- Label electrodes as anode and cathode and identify their polarities in voltaic cells and electrolytic cells, based on the type of reaction occurring at the electrode.
- Explain the direction of electron flow from anode to cathode in the external circuit, and ion movement across the salt bridge.
- Deduce the reactions of the charging process from given electrode reactions for discharge, and vice versa.
- Explain how current is conducted in an electrolytic cell.
- Deduce the products of the electrolysis of a molten salt.
- Deduce equations to show changes in the functional groups during oxidation of primary and secondary alcohols, including the two-step reaction in the oxidation of primary alcohols.
- Deduce equations to show reduction of carboxylic acids to primary alcohols via the aldehyde, and reduction of ketones to secondary alcohols.
- Deduce the products of the reactions of hydrogen with alkenes and alkynes.

Higher level (HL)

- Interpret standard electrode potential data in terms of ease of oxidation/reduction.
- Predict whether a reaction is spontaneous in the forward or reverse direction from $E^\ominus$ data.
- Determine the value for $\Delta G^\ominus$ from $E^\ominus$ data.
- Deduce from standard electrode potentials the products of the electrolysis of aqueous solutions.
- Demonstrate that electrolytic processes include the electrolysis of water and of aqueous solutions.
- Describe the effects of concentration and the nature of the electrode for the electrolysis of NaCl(aq) and $\text{CuSO}_4\text{(aq)}$.
- Deduce equations for the electrode reactions during electroplating.

Practical skills

Once you have completed this subtopic, apply your knowledge of electron transfer reactions by going to:

- [Practical 9: Investigating factors affecting electrochemical cells \(/study/app/preview-p/sid-426-cid-753301/book/investigating-factors-affecting-electrochemical-cells-id-46726/\)](#)
- [Practical 10a: Oxidation of alcohols \(/study/app/preview-p/sid-426-cid-753301/book/oxidation-of-alcohols-10a-id-46727/\)](#)
- [Practical 10b: Redox titration \(/study/app/preview-p/sid-426-cid-753301/book/redox-titration-10b-id-46728/\)](#).